

Exact Small- n Ordered and Classical Ruzsa–Szemerédi Densities

Task B1 technical note, v0.1 — companion to the execution plan and to `ors_lab`

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Abstract

Task B1 of the execution plan asks for a verified search package computing exact ordered/classical Ruzsa–Szemerédi densities $\text{ORS}_r(n)$ and $\text{RS}_r(n)$ at small n , with independent cross-validation, machine-checked certificates, and a targeted hunt for small- n ordered-vs-unordered separations. This note records the package `ors_lab` and its first results. Every value is produced by up to three *independent* backends (a dependency-free branch-and-bound search, a CNF/SAT encoding solved by PySAT, and an OR-Tools CP-SAT model), which must agree on the exact value, and every witness is accepted only after passing a separate $O(tn^2)$ verifier that shares no code with the solvers. The headline finding [P]: *ordered-vs-unordered separations exist already at $n = 7$ — $\text{ORS}_2(7) = 5 > \text{RS}_2(7) = 4$ and $\text{ORS}_2(8) = 8 > \text{RS}_2(8) = 6$ — with explicit witnesses that are valid ordered decompositions but provably *not* RS decompositions.* This is fully consistent with Pratt’s SOSA 2026 theorem that the two densities coincide *polynomially* (his equivalence is asymptotic; constant-factor gaps at small n are exactly what one expects), and it gives concrete extremal seeds for the WS-B construction effort. We also tabulate the linear / near- $n/2$ regime — the one that controls the AKK25 update time — where $\text{RS} = \text{ORS} = 1$ as soon as $2r$ approaches n , and where the two densities coincide on every cell we can pin exactly.

1 Definitions and the computational problem

We use the definitions of the execution plan (Def. 2.3), which follow Behnezhad–Ghafari [1]. [V]

Definition 1 (RS / ORS decompositions). *Let $M_1, \dots, M_t$ be pairwise edge-disjoint matchings on $[n]$, each of size exactly r , and let $G = M_1 \cup \dots \cup M_t$ be their union; write $V(M_i)$ for the set of vertices saturated by M_i . The sequence is an RS decomposition with parameters (r, t) if every M_i is an induced matching of G : for all $j \neq i$, no edge of M_j has both endpoints in $V(M_i)$. It is an ORS decomposition (for this order) if every M_i is induced in the suffix $M_i \cup M_{i+1} \cup \dots \cup M_t$: the condition is required only for $j > i$. We set $\text{RS}_r(n) = \max\{t : \text{an RS decomposition exists}\}$ and $\text{ORS}_r(n) = \max\{t : \text{an ORS decomposition exists}\}$.*

Every RS decomposition is an ORS decomposition under any order, so $\text{ORS}_r(n) \geq \text{RS}_r(n)$ for all n, r .

Encoding as an edge colouring. Computing $\text{RS}_r(n)$ and $\text{ORS}_r(n)$ is the optimisation “maximise t such that the pairs of $[n]$ admit a partial colouring $c : \binom{[n]}{2} \rightarrow \{0, 1, \dots, t\}$ with each colour class $i \geq 1$ a size- r matching satisfying the (ordered) induced condition.” Introducing, for each vertex w and colour i , the indicator $y_{w,i} = \bigvee_{p \ni w} x_{p,i}$ that w is saturated by colour i , the induced

condition is the clause family $x_{p,j} \Rightarrow \neg(y_{u,i} \wedge y_{v,i})$ for $p = \{u, v\}$ and $j \neq i$ (RS) resp. $j > i$ (ORS). This is the common skeleton of the SAT and CP-SAT encodings (§2).

2 Methodology: three independent backends and a verifier

Per the standing methodology of this research program (every reported number is computed at least two independent ways), `ors_lab` ships three solver backends and one independent certificate verifier.

Backend	Paradigm	Role
<code>search</code>	branch-and-bound, no dependencies	always available (CI / no network); colour-by-colour with incremental induced checks
<code>sat</code>	CNF + CDCL (PySAT/Glucose)	independent encoding; cardinality via sequential counters
<code>cpsat</code>	CP-SAT (OR-Tools)	independent encoding; the workhorse on the hard small- r cells

Symmetry breaking. All three backends pin M_1 to the canonical matching $\{(0, 1), (2, 3), \dots, (2r-2, 2r-1)\}$; since both the RS and ORS properties are invariant under relabelling the vertices, this is without loss of generality. The `search` backend additionally orders the remaining classes by their smallest edge, a complete symmetry break for RS (where colour classes are interchangeable) that is omitted for ORS (where the order is semantic).

Certificate verifier (the trust anchor). The value reported for a cell is accepted only if (i) at least one backend proves the matching decision and (ii) the witness passes `orslib.core.verify_decomposition`, an $O(tn^2)$ routine that recomputes $V(M_i)$ and the forbidden-pair condition from scratch and shares no code with any solver. As a built-in sanity check, every *ordered* witness that beats the corresponding RS value is additionally run through the *unordered* verifier and must *fail* it — confirming the gap is genuinely about ordering and not a solver artefact.

Exactness and budgets. A value is [P] (“proven”) when at least one backend certifies feasibility at t and infeasibility at $t + 1$ (or t equals the edge-count ceiling $\lfloor \binom{n}{2}/r \rfloor$). When the infeasibility proof exceeds the configured solver budget we report the verified lower bound $\geq t$ instead, tagged [E].

3 Results

3.1 The exact grid

Table 1 gives the exact values over the full (n, r) grid up to $n = 9$; each entry is $RS_r(n)/ORS_r(n)$, with bold marking a separation and $\geq$ a verified lower bound. The fragment is generated verbatim by `run_all.py`.

3.2 Separations exist at small n

Observation 1 (Small- n ordered-vs-unordered separations). $ORS_2(7) = 5 > RS_2(7) = 4$ and $ORS_2(8) = 8 > RS_2(8) = 6$ are proven (exact) separations; $ORS_2(9) \geq 11 > RS_2(9) = 9$ is a

$n \setminus r$	$r = 2$	$r = 3$	$r = 4$
4	1/1	.	.
5	1/1	.	.
6	3/3	1/1	.
7	4/5	1/1	.
8	6/8	1/1	1/1
9	9/ $\geq$ 11	3/3	1/1

Table 1: Exact $RS_r(n)$ / $ORS_r(n)$ (bold: a separation; $\geq$: verified lower bound).

n	r	$RS_r(n)$	$ORS_r(n)$	gap	status
7	2	4	5	1	proven
8	2	6	8	2	proven
9	2	9	≥ 11	2	lower-bound

Table 2: Small- n ordered-vs-unordered separations found by exhaustive search.

certified lower-bound separation. The witnesses below are valid ordered decompositions and fail the unordered RS condition.

$ORS_2(7) = 5 > RS_2(7) = 4$. $M_1 = \{\{0, 1\}, \{2, 3\}\}$
 $M_2 = \{\{0, 4\}, \{1, 5\}\}$
 $M_3 = \{\{0, 6\}, \{2, 4\}\}$
 $M_4 = \{\{1, 6\}, \{3, 5\}\}$
 $M_5 = \{\{2, 5\}, \{3, 4\}\}$

$ORS_2(8) = 8 > RS_2(8) = 6$. $M_1 = \{\{0, 1\}, \{2, 3\}\}$
 $M_2 = \{\{4, 6\}, \{5, 7\}\}$
 $M_3 = \{\{1, 6\}, \{3, 7\}\}$
 $M_4 = \{\{2, 6\}, \{3, 5\}\}$
 $M_5 = \{\{0, 7\}, \{1, 4\}\}$
 $M_6 = \{\{0, 5\}, \{3, 4\}\}$
 $M_7 = \{\{1, 5\}, \{2, 4\}\}$
 $M_8 = \{\{0, 6\}, \{2, 7\}\}$

These do *not* contradict Pratt [3]: his theorem says polynomial ORS density implies polynomial RS density at (slightly smaller) linear matching size, an *asymptotic* statement that is silent about, and fully compatible with, constant-factor gaps at small n . The practical value of the witnesses is twofold: they are exact extremal objects that seed the WS-B construction search (Task B2), and they make precise how much the ordering relaxation can buy in the regime where exhaustive certification is possible. [P]

3.3 The linear / near- $n/2$ regime

The regime $r = \Theta(n)$ is the one that controls the AKK25 update time. Table 3 records $r = \lfloor n/2 \rfloor - j$ for $j \in \{0, 1, 2\}$ up to $n = 20$; these cells have small t and are therefore cheap and exact

$n \setminus r$	$r = 2$	$r = 3$	$r = 4$	$r = 5$	$r = 6$	$r = 7$	$r = 8$	$r = 9$	$r = 10$
6	3/3	1/1	.	.	.	.	.	.	.
7	4/5	1/1	.	.	.	.	.	.	.
8	6/8	1/1	1/1	.	.	.	.	.	.
9	9/ ≥ 11	3/3	1/1	.	.	.	.	.	.
10	.	5/5	1/1	1/1	.	.	.	.	.
11	.	5/ ≥ 6	1/1	1/1	.	.	.	.	.
12	.	.	3/3	1/1	1/1	.	.	.	.
13	.	.	3/3	1/1	1/1	.	.	.	.
14	.	.	.	1/1	1/1	1/1	.	.	.
15	.	.	.	3/3	1/1	1/1	.	.	.
16	.	.	.	.	1/1	1/1	1/1	.	.
17	.	.	.	.	1/1	1/1	1/1	.	.
18	.	.	.	.	.	1/1	1/1	1/1	.
19	.	.	.	.	.	1/1	1/1	1/1	.
20	.	.	.	.	.	.	1/1	1/1	1/1

Table 3: The near- $n/2$ (linear matching size) regime $r = \lfloor n/2 \rfloor - j$.

even at larger n .

Two regularities are visible and exact [P]: (i) once $2r$ is within 1 of n the density collapses to $RS_r(n) = ORS_r(n) = 1$ (a second size- r matching would have an edge inside the near-spanning $V(M_1)$); and (ii) *on every cell of this regime that we pin exactly, ordered and unordered densities coincide* — the separations we found all live at the small- r (sub-linear) end, consistent with the intuition that ordering helps most when each matching saturates few vertices and the suffix relaxation removes many constraints. Figure 1 plots the $r = 2$ column.

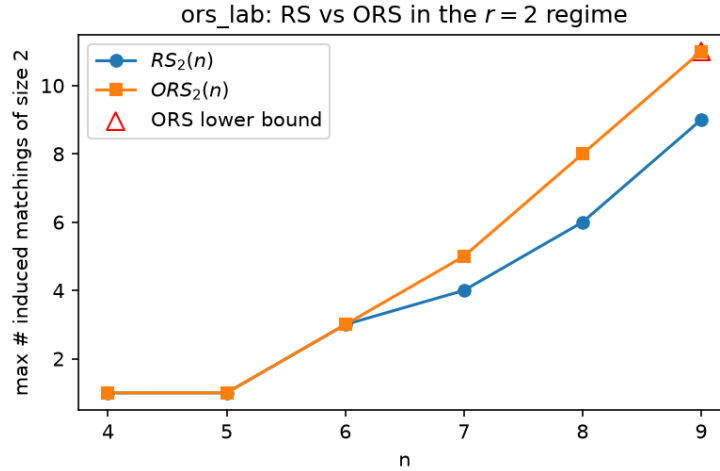


Figure 1: $RS_2(n)$ and $ORS_2(n)$ from `ors_lab`; open red markers are verified lower bounds (exact value not pinned within the solver budget).

4 Relation to the plan and next steps

This note delivers the second pre-committed fallback publishable unit (plan §7.2) and the `ors_lab/` code deliverable of WS-D, and it feeds WS-B directly. Concretely:

- **B0/B1 (frontier table).** The exact grid and the near- $n/2$ table are the small- n anchor of the (r, t) frontier the plan asks B0 to compile; the values are now machine-checked rather than quoted.
- **B2 (constructions).** The proven separation witnesses (§3.2) are exact extremal objects to lift; the targeted search can be re-pointed at the $r = \delta n$ regime as seeds.
- **L4 framing (WS-C).** Any exact $RS_r(n)$ value is an unconditional *upper bound on the AKK25 runtime at that (n, r)* ; the table is the small- n end of that bound.
- **Scaling.** The hard cells are the small- r ones; pushing toward the plan’s $n \lesssim 40$ aspiration is a backend swap (kissat / Gurobi) and a longer time budget, not a re-encoding — the three encodings are written to the same interface and remain cross-validated.

Status. Done: definitions [M]; three backends + verifier, cross-validated on the battery; exact grid to $n = 9$ and near- $n/2$ regime to $n = 20$ [P]; the $n = 7, 8$ separations [P] and the $n = 9, 11$ lower-bound separations [E]. Open within this lab’s budget: the exact values of the small- r ORS cells for $n \geq 9$.

References

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- [7] `audit_lab` / A0 technical note. Companion package and note, Task A0 (randomness audit), June 2026.